



Design and simulation of a novel 4H-SiC LGAD timing device

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Abstract

Purpose Silicon-based fast timing detectors have been widely used in high-energy physics, nuclear physics, space exploration and other fields in recent years. However, silicon detectors often require complex low-temperature systems when operating in irradiation environment, and their detection performance decreases with the increase in the irradiation dose. Compared with silicon, silicon carbide (SiC) has a wider band gap, higher atomic displacement energy, saturated electron drift velocity and thermal conductivity. Simultaneously, the low-gain avalanche detector avoids cross talk and high noise from high multiplication due to its moderate gain, and thus can maintain a high detector signal without increasing noise.

Aim Thus, the 4H-SiC particle detector, especially the low-gain avalanche detector, has the potential to detect the minimal ionizing particles under extreme irradiation and high-temperature environments.

Method In this work, the emphasis was placed on the design of a 4H-SiC low-gain avalanche detector (LGAD), especially the epitaxial structure and technical process which play main roles. In addition, a simulation tool—RASER (Radiation SEmiconductor)—was developed to simulate the performances including the electrical properties and time resolution of the 4H-SiC LGAD we proposed.

Conclusion The working voltage and gain effectiveness of the LGAD were verified by the simulation of electrical performances. The time resolution of the LGAD is (35.0 ± 0.2) ps under the electrical field of -800 V, which is better than that of the 4H-SiC PIN detector.

Keywords Silicon carbide · LGAD · Radiation-resistant · Time resolution

Introduction

Low-gain avalanche detector (LGAD), as a novel detector technology, has attracted extensive attention in recent years. By adding a gain layer under the cathode of the PIN detector, a collision–ionization multiplication of the electron–hole pair occurs in the gain layer, resulting in excellent timing res-

olution [1–10]. In particular, for the requirements of the High Granularity Timing Detector (HGTD) in ATLAS and the End-cap Timing Layer (ETL) in CMS experiments, silicon-based LGAD with 50 ps time resolution has been developed which can resist the radiation of $2 \times 10^{15} n_{eq}/cm^2$ [11–14]. However, it has been reported [15] that the charge collection

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efficiency (CCE) and time resolution of silicon-based LGAD will deteriorate with increasing irradiation flux and will not work normally under irradiation above $2 \times 10^{15} n_{eq}/cm^2$. In addition, in order to reduce power dissipation and improve the signal-to-noise ratio (SNR) of semiconductor detectors, silicon-based LGAD devices need to operate at a low temperature of $-30^\circ C$ [16, 17], which greatly increases the cost. Moreover, the cooling system not only increases volume of the detector system, but also increases the difficulty of maintaining the stability of the detector system. Therefore, there is a high demand for research into a semiconductor detector with strong radiation resistance, excellent time resolution and stable operation at room temperature or even high temperatures.

Silicon carbide (SiC), as the third-generation semiconductor, has been investigated as a detector material for nearly 20 years [18]. In recent years, SiC has received more attention due to the interest of semiconductor industries in the renewable energy revolution, leading to a broader application in power-efficient transistors, photovoltaic inverters and electric vehicle drive trains. Compared with silicon, SiC has higher band gap and atomic displacement, which leads to its strong potential for irradiation resistance [19]. In addition, the higher-breakdown electric field, saturated electron drift velocity and thermal conductivity mean that a SiC detector would have a faster time response and lower temperature sensitivity [20]. Zhang's group reported a 4H-SiC detector with a time response of 117 ps to alpha particles [21]. However, there are very few reports on the time response of minimum ionizing particles (MIPs). We have investigated the time performance of a 4H-SiC PIN detector with a thickness of 100 μm , and the time resolution of MIPs is 94 ps [22]. The average ionization rates of electron-hole pairs of MIPs in 4H-SiC and Si are 55 pairs/ μm and 75 pairs/ μm , respectively, which means that the response of the 4H-SiC detector is less than that of the Si detector. At present, the main problem of Si-based LGAD is the reduction in the timing performance caused by high-dose irradiation. Once 4H-SiC can be utilized to fabricate LGAD of high time resolution, combined with its irradiation resistance and temperature stability, 4H-SiC LGAD will be a candidate for high pileup, high-radiation environments and stable operation at room temperature.

In this work, a novel 4H-SiC LGAD timing device was designed by optimizing the doping concentration and epitaxial structure using the Radiation SEmiconductor (RASER) software, and its electrical characteristics were simulated. The design of the technical process has also been carried out, and RASER is further used to simulate the time resolution of the 4H-SiC LGAD. The time resolution of the 4H-SiC LGAD designed by adjusting the thickness and doping concentration is basically the same as that of the Si-based LGAD detector (34 ps) [23], which is obviously superior to the corresponding PIN device [22].

Design scheme of 4H-SiC LGAD

The design scheme of 4H-SiC LGAD will be introduced from two aspects: epitaxial structure and technical process.

Epitaxial structure of 4H-SiC LGAD

In this work, we design a 4H-SiC LGAD timing device by full epitaxial growth. Different from Si LGAD, the hole multiplication rate in 4H-SiC is greater than that of electrons. To achieve better carrier multiplication, the *N*-type epitaxial layers including bulk layer (N^-) and gain layer (N^+) have been selected. In detail, due to the high average ionization energy of the 4H-SiC, a relatively thicker bulk layer should be required to generate more initial ionized electron-hole pairs for incident particles, which can be read out by electronic readout system after undergoing 10 times of low-gain multiplication ($> 4fC$). Therefore, the thickness of the bulk layer of the 4H-SiC LGAD should be at least 50 μm . In addition, the doping concentration of the bulk layer with the thickness of 50 μm should be as low as possible in order to deplete the whole bulk region at operating voltage. At present, the epitaxial growth of 4H-SiC with a doping concentration of $10^{14} cm^{-3}$ can be realized in industry.

To achieve low-gain multiplication of 4H-SiC LGAD devices, an N^+ gain layer should be epitaxially grown above the bulk layer. Under the reverse bias, a narrow region of high electric field can be generated in the gain layer, causing the electron-hole pairs to be multiplied. Based on the experience of Si LGAD, a thickness of the gain layer is from 0.1 μm to 1 μm . In this work, 1 μm thickness gain layer with better uniformity has been selected. Meanwhile, the doping concentration range is $1.4 \times 10^{17} cm^{-3} \sim 1.48 \times 10^{17} cm^{-3}$, which restrict the corresponding multiplication coefficient and meet the industrial epitaxial process requirements. In addition, the P^{++} layer with a doping concentration of $5 \times 10^{19} cm^{-3}$ and a thickness of 0.3 μm should be epitaxially grown on the N^+ gain layer. The highly doped P^{++} layer is easy to form ohmic contact, improving its charge collection efficiency and time resolution.

Overall, the epitaxial structure of 4H-SiC LGAD is $P^{++}/N_{gain}^+/N_{bulk}^-/N_{buffer}/N_{substrate}^{++}$ as shown in Fig. 1a. The P^{++} layer is doped with aluminum ions and used as the anode, while the $N_{substrate}^{++}$ layer is doped with nitrogen ions and used as the cathode. The N_{gain}^+ layer is the gain layer, and its doping concentration and thickness play a decisive role in the gain, and affect the to be depleted voltage and breakdown voltage. The N_{bulk}^- layer is the bulk layer, the thickness of which dominates the generation of initial e-h pairs, which is expected to be depleted at low voltage. In order to prove the technological feasibility of the design scheme and the

accuracy of epitaxial parameters such as doping concentration and thickness, secondary ion mass spectrometry (SIMS) has been carried out for characterization at a depth of 3 μm , and the results are shown as Fig. 1b. The actual thickness and doping concentration of the device were consistent with the corresponding designed parameters, which provided support for our subsequent research on the time resolution of 4H-SiC LGAD.

Technical processes

It has been reported that the breakdown voltage of conventional planar 4H-SiC devices is mainly determined by the local electric field caused by the device boundary effect [24]. For the 4H-SiC LGAD designed in this work, the high-electric-field region generated in the gain layer is close to the device surface, and the boundary effect is particularly obvious, resulting in more serious premature breakdown. In addition, the operating voltage of the 4H-SiC LGAD needs to be greater than the full depletion voltage to work properly, which means that the operating voltage should be relatively high. Therefore, ensuring that the device does not break down in advance is a major consideration for the technical process. Etching the terminal junction is an effective method to avoid premature breakdown caused by the boundary effect of high electric field [25]. In this work, etching table with 7° has been used and the technical process designed on the existing epitaxial wafer is shown in Fig. 2a. In addition, the design and preparation of *P*-type ohmic contact electrodes are also very important to improve the performance of 4H-SiC LGAD. High-temperature anneal alloy based on Ni/Ti/Al has been widely used to produce *P*-type ohmic contact [26, 27], with specific contact resistance up to $1.8 \times 10^{-5} \Omega \text{ cm}^2$. The same alloy is used to make the *P*-type ohmic contact electrodes and *N*-type ohmic contact electrodes, and the annealing temperature is 800°C [28]. A low-resistivity ohmic contact can be formed between the metal and silicon carbide. The anode is N^{++} layer, and the cathode is P^{++} layer. Finally, in order to prevent gas from oxidizing the wafer, a layer of silica has been deposited as a passivation on the exposed part of electrodes. By comparing the thickness of silica and the laser wavelength, the silica with a thickness of 364 nm is finally determined. It not only provides protection, but also uses as a transparency-enhancing film for subsequent laser tests. The first version of the 4H-SiC LGAD device is named SICAR (Silicon CARbide) by our group.

To implement the above process, three masks has been designed, which are shown in Fig. 2b. The design consists of 8 structures (SICAR1-1–8), with further specific details provided in Table 1. Mask 1 is used to determine the size and pattern of etching table structure, Mask 2 is used to determine the *p*-type electrodes and Mask 3 is used to determine the electrode pads of the 4H-SiC LGAD. To

Table 1 The layout mask contains the device structure

Label	Type	Radius [μm]
SICAR1-1	Single	500
SICAR1-2	5×5	100
SICAR1-3	Single	100
SICAR1-4	2×2	100
SICAR1-5	2×2	100
SICAR1-6	Single	100
SICAR1-7	Single	200
SICAR1-8	Single	500

investigate the effects of size and shape on the electrical performance and time resolution, these 8 devices with different sizes and corner radii has been designed. Specifically, the shape and electrodes of SICAR1-1 ($5000 \times 5000 \mu\text{m}^2$) refer to the existing 4H-SiC PIN detector [22]. SICAR1-2 ($1000 \times 1000 \mu\text{m}^2$), a 5×5 array of SICAR1-6 is designed to improve the position resolution. Moreover, in order to study the multiplication heterogeneity of the 4H-SiC LGAD, a laser incidence hole with radius of (25 μm) is reserved on SICAR1-3 ($1000 \times 1000 \mu\text{m}^2$). Devices of SICAR1-4 ($1000 \times 1000 \mu\text{m}^2$) and SICAR1-5 ($1000 \times 1000 \mu\text{m}^2$) are 2×2 arrays with array spacing of 50 μm and 100 μm , respectively, which are designed to study the size of inactive region under different array spacing. In order to study the influence of corner radius on leakage current, 100 μm , 200 μm and 500 μm corner radius are designed for SICAR1-6 ($1000 \times 1000 \mu\text{m}^2$), SICAR1-7 ($1000 \times 1000 \mu\text{m}^2$) and SICAR1-8 ($1000 \times 1000 \mu\text{m}^2$) devices, respectively.

RASER

RASER (RAdition SEmiconductorR) is a software that simulates the electrical properties and time resolution of detectors made of Si and SiC materials. It was implemented in the ROOT framework using Python and relies on various open-source software such as DEVSIM, ROOT and Geant4. We utilize DEVSIM to simulate the electrical characteristics such as current–voltage and capacitance–voltage characteristics. Geant4 is used to simulate the process of electron–hole pair generation when incident particles pass through the detector. To estimate the *I*–*V* and *C*–*V* characteristics and time resolution performance of SiC detectors, a simulation tool–RASER is developed [29].

Firstly, based on an open-source software, DEVSIM is used to simulate the electrical performance including *I*–*V* and *C*–*V* characteristics. Different from the commercial simulation software, DEVSIM has strong expandability and is easy to be integrated with Geant4 and other software for

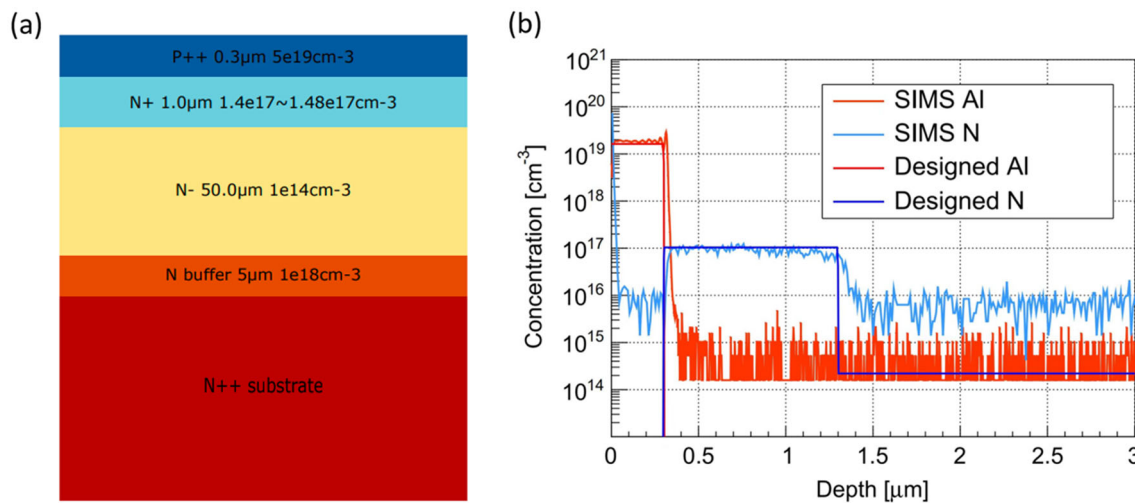


Fig. 1 **a** Schematic diagram of SiC LGAD epitaxial structure with doping concentration and thickness. **b** Comparison of SIMS measurement of doping concentration and thickness and the designed parameters

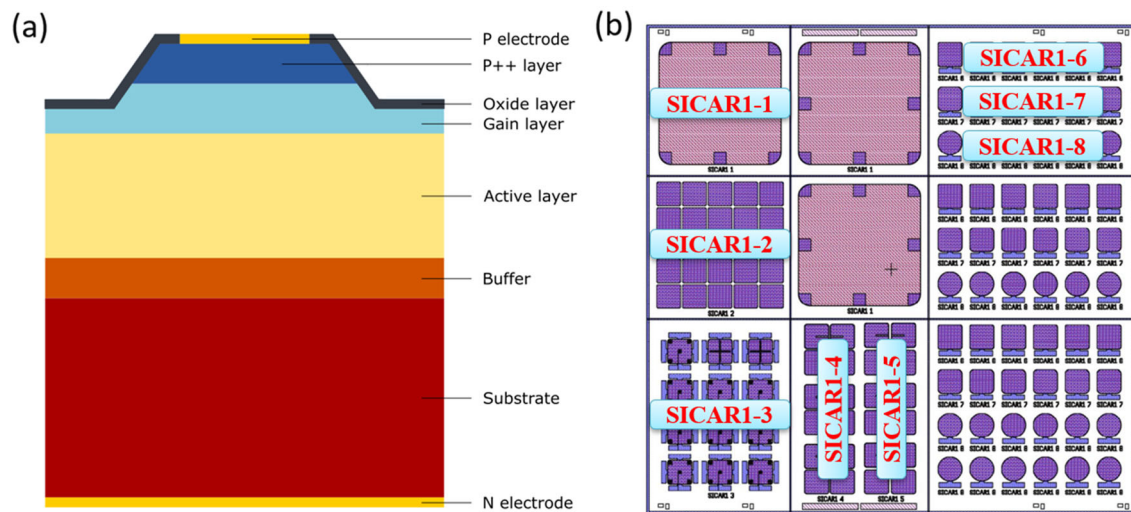


Fig. 2 **a** Sketch of the structure of an LGAD (not to scale). **b** SiC LGAD $2 \times 2 \text{ cm}^2$ layout mask. Different size and different rounded corner radius pads were allocated and aligned in this mask

detector simulation. But the DEVSIM is in the primitive development, and all the finite element equations should be written by users. As a TCAD device simulation package written in C++, with a Python front end, DEVSIM [30] is capable of simulating 1D, 2D and 3D structures with models describing advanced physical effects and uses the control volume approach for assembling partial differential equations (PDE's) on the simulation mesh.

Secondly, RASER based on the ROOT and Geant4 has been used to simulate the time resolution of 4H-SiC PIN [22] and 3D 4H-SiC detector [31]. In this work, RASER is also used to simulate the time resolution of 4H-SiC LGAD. When an incident-charged particle passes through the detector, it deposits a portion of its energy, producing electron-hole pairs. The electron-hole pairs move toward the electrodes

in response to the electric field, creating an induced current at the electrodes. This current can be calculated using Shockley-Ramo's theorem [32].

The simulation of 4H-SiC PIN detector

In order to demonstrate the reliability of the RASER simulation tool, the 4H-SiC PIN detector [22] is selected as an example to compare the simulation results with the experimental results. Firstly, the size of the 4H-SiC PIN device under investigation is $5 \text{ mm} \times 5 \text{ mm}$. It has an active epitaxy layer with a thickness of $100 \mu\text{m}$ and a doping concentration of $5.2 \times 10^{13} \text{ cm}^{-3}$, and both of the top and bottom electrodes are ohmic contacts. The comparison of C - V performance between experimental result and the DEVSIM simulation

Fig. 3 Comparisons between DEVSIM simulations and experimental results: **a** C - V performance of DEVSIM simulation and experimental results. **b** I - V performance of DEVSIM simulation and experimental result

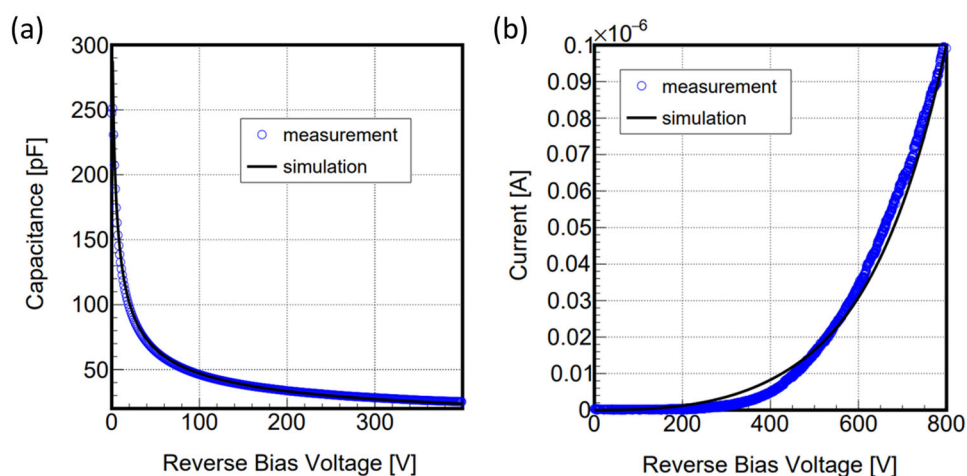
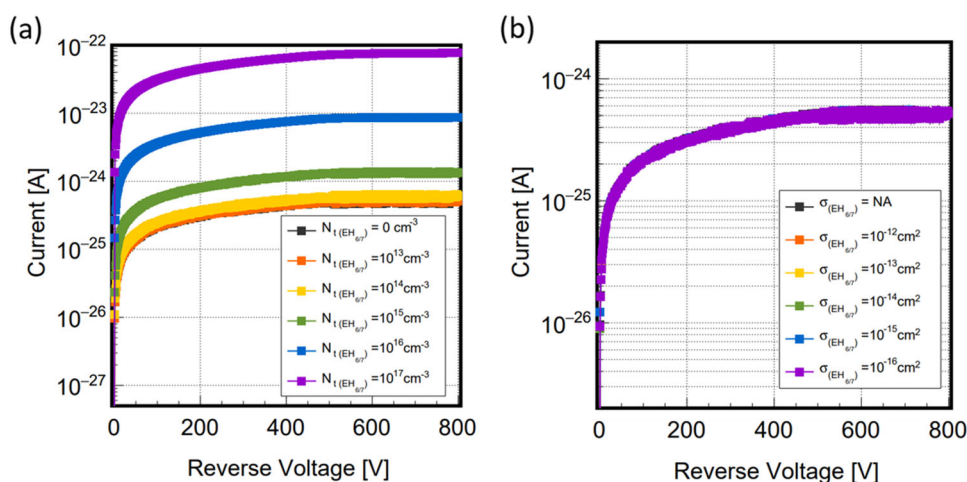


Fig. 4 **a** Effect of different $EH_{6/7}$ defect concentrations on leakage current. **b** Effect of capture cross section on leakage current



result is shown in Fig. 3a. The good agreement between simulation result and experimental result indicates that the geometry and doping input added in the DEVSIM simulation are correct. Furthermore, for the I - V simulation of the 4H-SiC PIN detector, considering the influence of tunneling effect, the Hurkx model [33] has been added into the generation and recombination (G&R) constant model, and the simulated results are closer to the experimental results. The band-to-band tunneling contribution R_{BTBT} can be obtained from the following expressions:

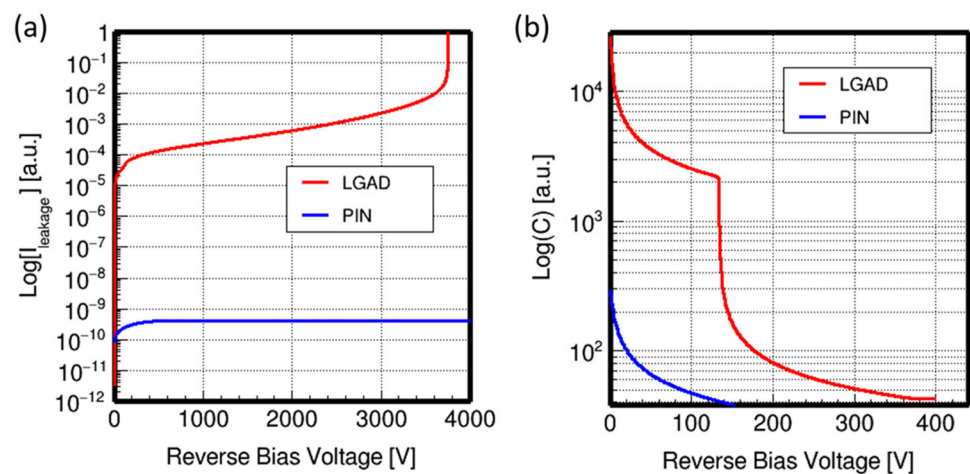
$$R_{BTBT} = -B|F|^{\sigma} D_{Kane} \exp \frac{-F}{F_0} \quad (1)$$

where F is the electric field strength (V/cm), and B , σ and F_0 are fitting parameters. Under the high electric field, the simulated results match well with the experimental results, but there are still significant differences under low electric field. This may be due to the complex defect features of SiC materials, which require further research. The coefficients of the Hurkx model are fixed by the fitting to the experimental results. The combination of I - V simulation results and

accurate C - V simulation results demonstrates that RASER is capable of basic device simulation and has a certain level of reliability.

In addition, the influence mechanism of some deep level defects on the leakage current was also studied. Two kinds of deep level defects, $Z_{1/2}$ and $EH_{6/7}$, in 4H-SiC materials were mainly investigated [34]. $Z_{1/2}$ and $EH_{6/7}$ are both electron traps in 4H-SiC material. $Z_{1/2}$ has a thermal ionization energy of approximately 0.43 eV $\sim$ 0.71 eV [35]. This means that at lower temperatures, $Z_{1/2}$ can trap electrons. It has been observed in electron-irradiated 4H-SiC samples and has a relatively large capture cross section. On the other hand, $EH_{6/7}$ has a higher thermal ionization energy of approximately 1.65 eV [36]. Compared to $Z_{1/2}$, $EH_{6/7}$ requires higher temperatures to trap electrons. $EH_{6/7}$ also has a larger electron capture cross section, indicating a higher efficiency in capturing electrons. For $Z_{1/2}$, the defect concentration ranges from $1 \times 10^{13} \text{ cm}^{-3}$ to $1 \times 10^{15} \text{ cm}^{-3}$, and the capture cross section ranges from $1 \times 10^{12} \text{ cm}^2$ to $1 \times 10^{16} \text{ cm}^2$. However, there is no significant effect on the leakage current from $Z_{1/2}$, which may be due to its relatively low energy level. On the

Fig. 5 **a** Current–voltage characteristics and the **b** capacitance–voltage characteristics of 4H-SiC LGAD and 4H-SiC PIN, used DEVSIM simulation



other hand, for $\text{EH}_{6/7}$ with an energy level of 1.25 eV to 1.73 eV, the leakage current increases slightly with the increase in the defect concentration shown as Fig. 4a. The change of the capture cross section has little effect on the leakage current shown as Fig. 4b. From the simulation, the deep energy level defect has a slight effect on the leakage current of 4H-SiC PIN device. Furthermore, it has been reported that macroscopic defects would greatly affect the carrier lifetime of the SiC devices, thus affecting the leakage current [37, 38]. Therefore, it is particularly important to avoid macroscopic defects in the preparation process of SiC detectors.

The simulation of 4H-SiC LGAD detector

RASER was used to simulate the I – V and C – V curves of the 4H-SiC LGAD. As shown in Fig. 5a, the breakdown voltage of this device can be obtained from the I – V curve, which is approximately 3700 V. The C – V curve, shown in Fig. 5b, demonstrates the depletion voltage of gain layer V_{GL} and the full depletion voltage V_{FD} . In detail, different from the PIN device, the C – V curve of the LGAD device shows a platform at around 130 V, which indicates that the gain layer of the device can be completely depleted at around 130 V. In addition, the full depletion voltage of the 4H-SiC LGAD obtained from Fig. 5b is 400 V. Combined with the breakdown voltage, it can be considered that the working voltage range of the 4H-SiC LGAD is 400–3700 V.

Figure 5a, b shows the I – V and C – V curves of LGAD and PIN, respectively. These two devices are identical in structure and doping, except that the PIN has no gain layer. In terms of the I – V curve, the breakdown voltage V_{BD} of 4H-SiC LGAD is about 3700 V, and it is smaller than that of PIN which does not breakdown until the reverse voltage is 4000 V. The reason for the phenomenon is that the high doping concentration of the gain layer in LGAD created an electric field peak, which causes multiplications of the carriers and can generate avalanche. At the same voltage, the leakage

current of LGAD is larger than that of PIN due to the presence of the gain layer in LGAD. However, in terms of the C – V curve, LGAD has one more turning point than PIN. Different from the PIN devices, with the increase in the reverse voltage, the gain layer of the LGAD device is depleted before the bulk layer.

The I – V and C – V curves of PIN and LGAD we simulated show that, unlike the structure of PIN, LGAD has a gain layer and low-gain multiplier effect. The structure of LGAD that we have designed is reasonable.

The time resolution simulation of 4H-SiC LGAD detector

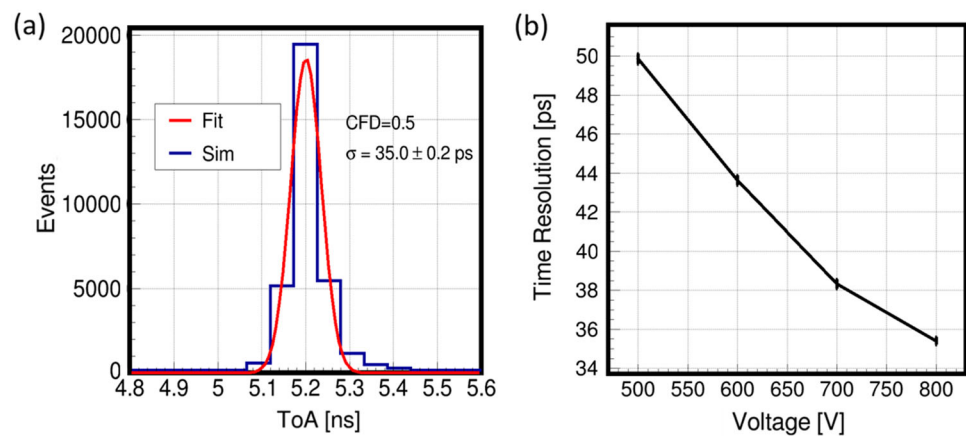
The time resolution of a detector is defined by the accuracy of the measured arrival time of a detected particle. It can be calculated by the following formula [39]:

$$\sigma_t^2 = \sigma_{\text{TimeWalk}}^2 + \sigma_{\text{Landau}}^2 + \sigma_{\text{Distortion}}^2 + \sigma_{\text{Jitter}}^2 + \sigma_{\text{TDC}}^2 \quad (2)$$

where the terms σ_{TimeWalk} and σ_{Landau} are from the random energy deposition of the detected particle, the physical process corresponding to which is simulated by Geant4. The term σ_{TimeWalk} , or the uncertainty of signal exceeding given threshold, can be eliminated by the constant fraction discrimination (CFD) method. The term $\sigma_{\text{Distortion}}$ is caused by the fluctuation of carrier velocity, which could be calculated from the Langevinian random motion of the carriers. Noise in the circuit contributes to the term σ_{Jitter} . Since we set the electronics the same as [22] in the simulation, a Gaussian noise, parameters of which measured from [22], is used. The σ_{TDC} is introduced by analog-to-digital converting and is often negligible.

We simulated 50,000 events of beta particle detection using the SiC LGAD detector under a -800 V bias voltage. The time of arrival (ToA) of the particles was then recorded in order to determine the time resolution. The obtained time

Fig. 6 Time resolution simulation results of the LGAD detector **a** when the reverse bias voltage is 800V and **b** the reverse bias voltage from 500V to 800V



resolution is (35.0 ± 0.2) ps, as shown in Fig. 6a. We also simulated beta particle detection under different detector working voltages. As figure 6b shows, the time resolution of the 4H-SiC detector decreases with the increase in the reverse voltage. This is due to the fact that the drift velocity of the carriers increases as the voltage increases, resulting in the carriers being collected at a faster rate.

Conclusion

In conclusion, the 4H-SiC LGAD has been designed including the epitaxial structure and technical processing and is currently in production. The electrical performance and time resolution of 4H-SiC LGAD have simulated by adjusting doping concentration and device structure using an open-source software DEVSIM. Combined its I - V with C - V simulation results, the device has a full depletion voltage of 400 V and a breakdown voltage of 3700 V, which means that its operating range is 400–3700 V. Furthermore, the corresponding simulation exhibited that the device has a time resolution of (35.0 ± 0.2) ps at a reverse bias voltage of 800 V, exceeding that of the SiC PIN detector. This suggests that the 4H-SiC LGAD detector has the potential to be used in high-irradiation environment at high temperature. This work paves the way for the future development of 4H-SiC LGAD and its application in an extreme environment.

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Declarations

Conflict of interest We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, and there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled “Design and simulation of a novel 4H-SiC LGAD timing device.”

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